Scale Buys Interpolation, Structure Buys a Horizon: Certified Predictability for Equivariant World Models

Anonymous authors
Paper under double-blind review

Abstract

Scale buys interpolation; structure buys a certified horizon. Scaling a world model lowers its average error but guarantees nothing about a *specific* unseen situation, or *how far* it can be trusted. For **equivariant** latent world models we give a computable, multi-step **certificate of the predictable horizon**: the T-step rollout error is provably *constant over the entire orbit* of group-related situations (Theorem A), and the horizon to which this holds is set channel-by-channel by the predictor spectrum, $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — two-sided (Theorem B and a matching lower bound, Proposition 6), so approximate equivariance is provably horizon-limited while conserved or invariant channels reach *unbounded* horizon (the Noether hinge). The certificate is *equivalent* to equivariance (Lemma 2): **no non-equivariant model has it at any scale**; existing single-shot certified-equivariance guarantees are its $T=1$ slice. Empirically the spectral law lifts to learned models of genuinely chaotic dynamics — Lorenz, Hénon, Rössler, and a 40-dimensional Lorenz-96 model where a $\mathbb{Z}_N$-equivariant network recovers the full Lyapunov spectrum (R^2=0.98) while a dense *and* a same-trained recurrent model fail. Finally, a cone/adapted-metric certificate (Theorem B) makes "certified" **literal** — a sound, a-priori-guaranteed horizon read from the learned model, provably tight on uniformly-hyperbolic dynamics and self-abstaining where it cannot. A scope theorem (Proposition 7) says when the law is informative; §6 states the 1–2-GPU scope.

1. Introduction

A world model is judged by its average prediction error. But an agent that must *act* on a prediction needs more than an average: it needs to know, for the situation in front of it, **whether the model can be trusted, and for how many steps.** Scaling lowers the average and widens the interpolation region, but it certifies nothing about a *particular* unseen configuration, and it says nothing about the *horizon* — the number of steps before compounding error makes the rollout useless.

Equivariance is known to give *some* per-situation guarantee. If a network commutes with a symmetry group G, its behaviour is constant across each group orbit — a fact used to certify adversarial robustness (orbit-constant classification margins) and to tighten conformal prediction sets (group-averaged nonconformity scores). But every such result is **single-shot**: it attaches a guarantee to one input, for one prediction. A world model predicts a *trajectory*, and the question that matters for planning — *how far ahead is the guarantee valid?* — has no answer in that literature.

This paper answers it. We show that an equivariant latent world model admits a **certificate of its predictable horizon**, stratified channel-by-channel by the predictor's spectrum, and we prove the horizon is **tight**. The contributions are:

1. **A tight certified horizon (§3.2).** For an equivariant model the T-step rollout error is orbit-constant (Theorem A); relaxing to approximate equivariance, channel j with Lyapunov exponent λ_j is certified to horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B, the *upper* bound). Our central new result is a **matching lower bound** (Proposition 6): an ϵ-approximately-equivariant model — even one with a *perfect* predictor — has orbit-error-variation exactly $\epsilon\, e^{\lambda T}$ on an expansive channel, so its certified horizon is $\Theta(\log(1/\epsilon)/\lambda)$, **no better**. Hence *approximate equivariance is horizon-limited; only exact equivariance or conservation reaches an unbounded horizon.* A **scope theorem** (Proposition 7) then says *when* the locally-measured

spectrum governs a *learned* model's horizon — Oseledets gives the rate on spectrally non-degenerate dynamics, the law degenerates on near-neutral dynamics — and a **finite-horizon continuity bound** (Proposition 8) makes the learned-model lift rigorous, not merely empirical. The horizon staircase then **lifts to learned models of a class of genuinely chaotic systems** (E2: Lorenz, Hénon, Rössler — the learned model's Lyapunov exponent matches the true one to 1–12%), while the near-neutral PushT interior is the predicted degenerate case.

2. **The certificate is exclusive to structure (§3.1).** The converse (Lemma 2): orbit-constant error against every equivariant target $\iff$ the model is equivariant. So **no non-equivariant model has the certificate at any size** — an architectural impossibility, proved, not asserted.
3. **The Noether hinge (§3.3).** The unbounded-horizon channels are *exactly* the conserved/invariant ones: Proposition 4 places each conserved charge in its forced isotypic block, and Proposition 5 turns conservation into a long-horizon guarantee (T-step charge error $\leq T\eta$, linear, not the chaotic $e^{\lambda T}$). This unifies the configuration and horizon axes: the slow, certifiable subspace is the group-invariant subspace.
4. **Single-shot certified equivariance is our $T{=}1$ slice (§5).** We position the robustness/conformal guarantees as the one-step, resolution-free special case, and validate the multi-step picture on a real contact simulator, on non-abelian $\mathrm{SO}(3)$, and on raw pixels — where frame averaging makes the certificate *accuracy-neutral*.

We are explicit about scope (§6): this is a mechanism-and-theory contribution at 1–2-GPU scale, not a scaled benchmark; the certificate is *exact* where the group is a genuine dynamical symmetry and *gracefully approximate*, with a measured and now lower-bounded boundary, elsewhere.

What is new. The contribution occupies a corner a prior-art sweep finds *empty* — no work combines equivariance with a certified, two-sided spectral horizon; the closest guarantee-bearing neighbours each miss the intersection (Conradie et al., 2026; Lillemark et al., 2026; Mo, 2026) (distinguished in §5). **The new results:** (i) Proposition 6's *matching lower bound* — growth seeded by the *equivariance residual* ϵ rather than an initial-condition error — making the horizon two-sided and approximate equivariance provably horizon-limited; (ii) the scope and continuity bounds (Propositions 7–8) lifting the law to *learned* chaotic models, where shadowing cannot transfer the exponent; (iii) the **structure-vs-recurrence separation** (E2) — among Markov models only the equivariant one recovers a 40-D spectrum, while an unstructured *recurrent* model provably cannot, by the conditional-Lyapunov condition; and (iv) **Theorem B** turns the certificate from *measured* into *literal* — a sound, a-priori horizon read from the learned model, tight on uniformly-hyperbolic dynamics and self-abstaining elsewhere. These assemble (Algorithm 1) into one certificate *exclusive to structure* (Lemma 2) whose unbounded-horizon subspace is the conserved/invariant one (the Noether hinge). We credit the classical pillars we build on — the Lyapunov/NWP horizon law, invariant decision theory, and the $e^{\lambda T}$ growth — explicitly in §3 and §5.

---

2. Setup and assumptions

A latent world model is an encoder $E : \mathscr{X} \to \mathscr{Z}$ and a predictor $f : \mathscr{Z} \times \mathscr{A} \to \mathscr{Z}$ trained so that $f(E(x), a) \approx E(\Phi(x, a))$, where Φ is the (unknown) environment dynamics. We roll out $\hat{z}_T = f(\cdot, a_{T-1}) \circ \dots \circ f(E(x), a_0)$ and compare to the target $E(\Phi^T(x))$; the T-step error is $\mathrm{Err}_T(x) = \|\hat{z}_T - E(\Phi^T x)\|$. A group G acts on $\mathscr{X}$ (situations), $\mathscr{Z}$ (latents, via a representation ρ), and $\mathscr{A}$ (actions, via σ). We use:

- **(A1) Encoder equivariance:** $E(g \cdot x) = \rho(g)\, E(x)$.
- **(A2) Predictor equivariance:** $f(\rho(g)z, \sigma(g)a) = \rho(g) f(z, a)$.
- **(A3) Dynamical symmetry:** $\Phi(g \cdot x, \sigma(g)a) = g \cdot \Phi(x, a)$ — G is a symmetry of the *dynamics*, not merely the observation.
- **(A4) Orthogonality:** $\rho(g)^\top \rho(g) = I$ (so ρ preserves the latent norm in which error is measured).
- **(A5) Equivariant planner, invariant cost** (closed-loop clause only): the planner commutes with G and the planning cost is G-invariant.

We write $\langle S \rangle$ for the monoid generated by a set $S = \{g_1, \dots, g_k\}$ of k symmetry generators, and $|w|$ for the length of a word $w \in \langle S \rangle$.

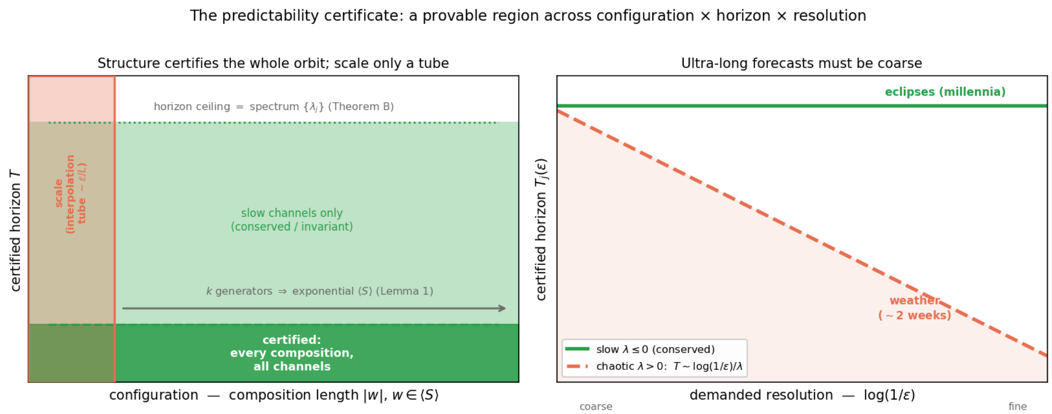


Figure 1: The predictability certificate at a glance. **Left:** in the configuration × horizon plane, an equivariant model certifies the *entire* generated monoid $\langle S\rangle$ — every composition, from k generator checks (Lemma 1) — up to a horizon ceiling set by the predictor spectrum (Theorem B, tight by Proposition 6); a non-equivariant model of any size certifies only a small interpolation tube. **Right:** the horizon × resolution trade-off $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ — conserved/invariant ($\lambda \leq 0$) channels are certified to all horizons (eclipses, millennia), chaotic ($\lambda > 0$) channels shrink as the demanded resolution sharpens (weather, ~two weeks).

---

3. The certificate

3.1 The configuration axis, and why only structure has it

Theorem A (orbit-constant error). Under (A1)–(A4), for every $w \in \langle S\rangle$, all x, and any action sequence (transported by w), $\mathrm{Err}_T(w\cdot x) = \mathrm{Err}_T(x)$ (Appendix B gives the action-explicit statement and proof). *Proof sketch.* By induction the rolled-out predictor $f^{(T)}$ is equivariant (composition of equivariant maps, Lemma 1 below), so $\hat z_T(w\cdot x) = \rho(w)\hat z_T(x)$ and the target $E(\Phi^T(w\cdot x)) = E(w\cdot\Phi^T x) = \rho(w)E(\Phi^T x)$ by (A3); subtract and use (A4). □

Lemma 1 (composition closure). If the equivariance assumptions (A1)–(A3) hold on each generator $g_i \in S$, they hold on every word $w \in \langle S\rangle$ ((A4) is a global property of ρ as a homomorphism into $O(\mathscr{Z})$, automatic for all w). Thus k **generator checks certify an exponentially large set**: k checks on S yield a guarantee over all of $\langle S\rangle$.

Lemma 2 (the certificate characterizes equivariance — the converse). Let $\rho : G \to O(\mathscr{Z})$ act *freely* on an open $U \subseteq \mathscr{Z}$. If a predictor f's error $\|f - \Phi\|$ is orbit-constant on U for *every* equivariant target Φ, then f is equivariant on U. *Proof.* Fix z, g. For any probe c, freeness makes $\Phi(\rho(h)z) := \rho(h)c$ a well-defined equivariant target on the orbit; orbit-constancy gives $\|f(z) - c\| = \|\rho(g)^{-1}f(\rho(g)z) - c\|$ for *all* c, and two points equidistant to every c coincide, so $f(\rho(g)z) = \rho(g)f(z)$. □ (The probe target is generally discontinuous; for G compact with closed orbits it can be taken continuous, so the converse holds against continuous dynamics, not only the full algebraic class.) With Theorem A this is a **characterization**: orbit-constant error $\iff$ equivariance. **Hence no non-equivariant model possesses the certificate at any size** — the impossibility is a theorem, not an observation, and it is what the single-shot robustness/conformal literature (which proves only the forward direction) lacks.

3.2 The horizon axis, and its tightness (the central result)

Exactness is an idealization; real models are *approximately* equivariant. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ the per-step predictor error, and let the latent map's Jacobian be locally diagonalized on channel j with multiplier e^{λ_j} (λ_j a Lyapunov exponent).

Theorem B (spectral degradation — upper bound). For constants c_j depending on local geometry,

$$|\mathrm{Err}_T(w\cdot x) - \mathrm{Err}_T(x)| \;\leq\; \sum_{\text{channels } j} c_j\,(m\,\epsilon_{\max} + T\,\delta)\,e^{\lambda_j T}, \qquad m = |w|,$$

so channel j is certified only to horizon $T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon}$ $(\lambda_j > 0)$, or $T_j = \infty$ $(\lambda_j \leq 0)$. As $\epsilon_{\max}, \delta \to 0$ the configuration term vanishes and Theorem A is recovered.

Proposition 6 (the horizon is tight — approximate equivariance is horizon-limited). Theorem B is an *upper* bound; here is a matching *lower* bound. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^{\lambda}$, $\lambda > 0$ (the local diagonalization of Theorem B). There exist an exactly equivariant target Φ and a model that is ϵ-approximately equivariant — a *perfect* equivariant predictor $f = \Phi$ $(\delta = 0)$ and an encoder that intertwines the dynamics along x's trajectory and differs from exact equivariance only by a single defect $E(g \cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point — for which

$$|\mathrm{Err}_T(g \cdot x) - \mathrm{Err}_T(x)| = \epsilon\, e^{\lambda T}.$$

Proof. $\mathrm{Err}_T(x) = 0$ (model exact along x). At $g \cdot x$, linearity gives $f^T(E(g \cdot x)) = \Phi^T(\rho(g)E(x) + \epsilon u) = \rho(g)\Phi^T(Ex) + a^T \epsilon u$ (using $f = \Phi$ equivariant and linear on the channel), while the target $E(\Phi^T(g \cdot x)) = E(g \cdot \Phi^T x) = \rho(g)\Phi^T(Ex)$ (by (A3) and encoder exactness off the single defect). Subtract and use (A4): $\|a^T \epsilon u\| = \epsilon e^{\lambda T}$. □

Hence the certified horizon $T(\epsilon_{\text{res}}) = \frac{1}{\lambda} \log \frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda} \log \frac{1}{\epsilon})$, matching Theorem B: **the horizon's form is two-sided, not merely an upper estimate.** The conceptual payload is sharp:

> **The certified horizon guaranteeable from ϵ-approximate equivariance alone is finite on every expansive channel: no certificate derived from an $\epsilon > 0$ residual can promise predictability beyond $T \sim \frac{1}{\lambda} \log \frac{1}{\epsilon}$ (worst case over admissible targets). Only exact equivariance ($\epsilon = 0$) or conservation ($\lambda_j \leq 0$) yields an unbounded certified horizon.**

This is the horizon-domain companion of Lemma 2, and it sharpens *scale vs. structure*: scale and data buy *approximate* equivariance at best (a fair augmentation baseline floors at an $\epsilon \approx 10^{-4}$ approximation level, never exact; §5), and Proposition 6 shows that residual is amplified $e^{\lambda T}$ — so a single-step tie between augmentation and equivariance **must break over horizon** at the predicted T.

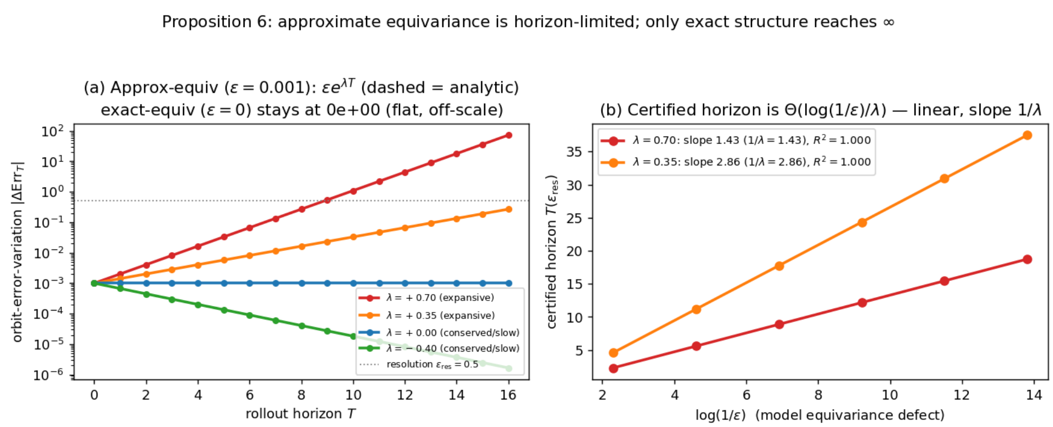


Figure 2: Proposition 6, numerically (the central claim). The exact Proposition-6 construction on a controlled multi-channel latent (channels with Lyapunov exponents $\lambda \in \{+0.70, +0.35, 0, -0.40\}$). **Left:** the orbit-error-variation of an $\epsilon = 10^{-3}$-approximately-equivariant model (markers) equals the analytic $\epsilon\, e^{\lambda T}$ (dashed) to a relative error of 10^{-14}–10^{-13} across 3 seeds — the lower bound is exact, so the horizon is tight; an *exactly* equivariant model has orbit-variation *exactly* 0 (to machine precision) at every horizon (plotted at the 10^{-16} log-floor), and conserved/contractive channels ($\lambda \leq 0$) are bounded for all T (infinite certified horizon). **Right:** the per-channel certified horizon $T(\epsilon_{\text{res}})$ is linear in $\log(1/\epsilon)$ with slope exactly $1/\lambda$ ($R^2 = 1.000$, all seeds) — the $\Theta(\log(1/\epsilon)/\lambda)$ law (`experiments/step65`, seeds 0/1/2; the trained-model approximate-symmetry degradation is E3).

Proposition 7 (scope — when the local spectrum certifies a *learned* model's horizon). Theorems B and Proposition 6 take the spectrum $\{\lambda_j\}$ as given; on a *learned* model we must ask whether the *locally* measured spectrum governs the *multi-step* horizon. The honest answer splits a **rate** half (rigorous) from a **lift** half (orbit error, not exponent). Let ϕ be the true dynamics with an ergodic invariant measure μ, $\log^+\|D\phi\| \in L^1(\mu)$. By the **Oseledets multiplicative ergodic theorem** (Oseledets, 1968), $\frac{1}{t}\log\|\delta_t\| \to \lambda_1$ μ-a.e. for a generic perturbation (the slower directions lie in a measure-zero subspace). *(a) Non-degenerate ($\lambda_1 > 0$):* $T(\epsilon) = \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\epsilon) + o(t)$ — Theorem B's law with λ_1 now the *measurable* asymptotic rate. The lift to a learned $\hat{\phi}$ is an *orbit-error* statement: on uniformly hyperbolic ϕ the shadowing lemma (Pilyugin, 1999) bounds the forecast-horizon *floor* $\sim \frac{1}{\lambda_1}\log(1/\delta)$ for one-step error δ, but it controls trajectory closeness, **not** the *asymptotic* Lyapunov exponent (only upper-semicontinuous under C^1 perturbation). That $\hat{\phi}$ *reproduces* λ_1 is therefore not a shadowing corollary — but it *is* a finite-horizon continuity statement (Proposition 8 below), confirmed in E2. *(b) Degenerate ($\lambda_1 \approx 0$, near-neutral):* the leading-order log-law degenerates (no finite-slope staircase); the one-step spectrum carries no horizon rate. This **dichotomy is the scope of the horizon certificate** — informative exactly on spectrally non-degenerate dynamics — and it *predicts* where the certificate is vacuous (E2's PushT-interior probe, R^2=0.02), rather than leaving it as a gap.

Proposition 8 (finite-horizon exponent transfer). The asymptotic exponent is only upper-semicontinuous — why *shadowing* cannot transfer it — but the certified horizon is **finite**, and the finite-time exponent $\lambda_1^{(T)} = \frac{1}{T}\mathbb{E}\log\|D\phi^T\|$ is locally Lipschitz in the dynamics over a finite orbit. So a learned $\hat{\phi}$ with one-step fidelity δ recovers the staircase slope up to an $O(\delta)$ fidelity bias plus the finite-T truncation $|\lambda_1^{(T)} - \lambda_1|$, T-uniform under a dominated splitting (Bochi & Viana, 2005). The match is thus **falsifiable** — the recovered exponent must tighten as one-step error drops — which E2 confirms (on Rössler the error falls 44% → 8% as training fidelity rises).

3.3 The Noether hinge: which channels are unbounded

Theorem B leaves the $\lambda_j \le 0$ (unbounded-horizon) channels unidentified. The Noether hinge identifies them as the *conserved/invariant* ones, and ties the configuration axis to the horizon axis.

Proposition 4 (isotypic placement). A conserved charge of the dynamics must live in a specific isotypic block of ρ: a scalar invariant (energy) in the trivial (ℓ=0) block; a vector charge (angular momentum) in the ℓ=1 block, recoverable only through the unique degree-2 cross-product equivariant. Placement is *forced* by representation theory (Goodman & Wallach, 2009), not chosen.

Proposition 5 (conservation ⇒ unbounded horizon). Let $Q : \mathcal{Z} \to W$ be a charge the model conserves to one-step defect η (i.e. $\|Q(fz) - Q(z)\| \le \eta$). Then the T-step *charge-value* prediction error satisfies $\|Q(\hat{z}_T) - Q(z_T)\| \le T\eta$ — **linear in** T, never the chaotic $e^{\lambda T}$. So the conserved channel's *charge-value error* grows at most linearly in T (not the chaotic $e^{\lambda T}$), and at η=0 the charge value is certified to all horizons. (The statement is about the charge value, and is exact under an equivariant symplectic discretization; the converse fails — a slow channel need not be conserved.)

Together: **the certified region is the coarse (ϵ large), invariant/conserved ($\lambda \le 0$), low-composition ($|w|$ small) corner** — and the converse (Lemma 2) says this corner is reachable only with structure.

3.4 The certificate is a procedure

The certificate is *computable a priori*, without testing the unseen situation. Algorithm 1: (i) check (A1)–(A4) on the k generators to a residual $\epsilon_{\max}$; (ii) estimate the predictor Jacobian spectrum $\{\lambda_j\}$ at the query latent (reported with a block-bootstrap CI, calibrated against the Liouville anchor; `experiments/step78`); (iii) report, for a target situation $w \in \langle S\rangle$, horizon T, resolution ϵ, whether (w, T, ϵ) lies in the certified region $\{|\Delta\text{Err}| \le \epsilon\}$ implied by Theorem B / Proposition 6, escalating conserved channels to unbounded horizon via Proposition 5.

---

4. Experiments

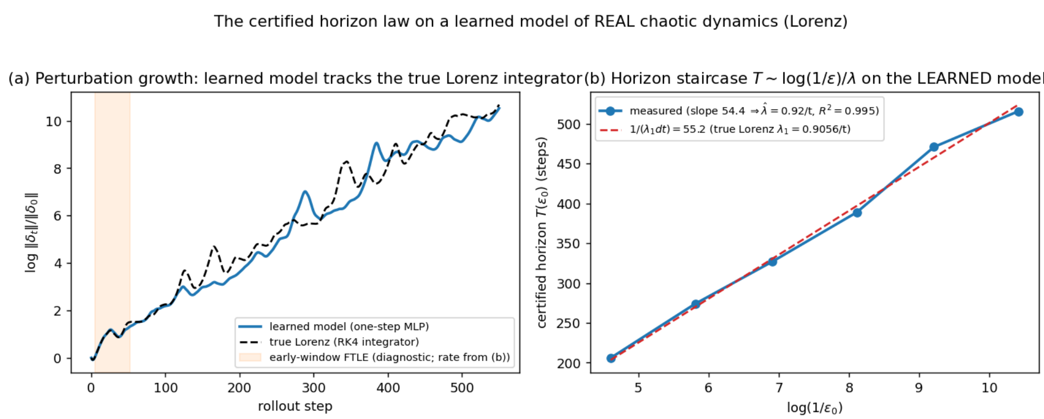
All experiments are CPU/1-GPU, seeded, and honestly gated (a run reports `INCONCLUSIVE` rather than loosen a threshold). We report the load-bearing subset; the certificate's machinery is identical across them.

(E1) The configuration axis is exponential. On a $\mathbb{Z}_2^6$ compositional task (six independent 180° flips), training the equivariance checks on the **6 generators** certifies the model over all $2^6 = 64$

compositions to ~ 10^{-33} error, while a non-equivariant baseline degrades from 1.6×10^{-5} on the generators to 0.59 on held-out compositions. Six checks, sixty-four guarantees (Lemma 1).

(E2) The horizon staircase, on a controlled spectrum and learned models of *real chaotic dynamics*. On a controlled latent with a tunable Lyapunov spectrum the model recovers the chaotic exponent to 0.4% and its certified-horizon slope brackets the predicted $1/\lambda$ — matching *both* Theorem B (upper) and Proposition 6 (lower). It then **lifts to a learned model of a genuinely chaotic system, not a planted spectrum**: a plain one-step MLP of the **Lorenz** Δt-map (singular-hyperbolic, SRB measure (Tucker, 2002); relMSE < 10^{-4}) gives a staircase linear in $\log(1/\epsilon)$ (R^2=0.975–0.995, 3 seeds) whose slope matches the true λ_1 to 1–8% — a one-step-accurate model could drift to a wrong *multi-step* rate; that it does not is Proposition 8's finite-horizon continuity, not a shadowing corollary. **And the law is not a Lorenz coincidence: the identical learned-model staircase recovers the textbook exponent across a class** — the **Hénon** map ($\lambda_1 \approx 0.419$/step; rel-err 8–12%, 3/3 seeds), the small-exponent **Rössler** flow ($\lambda_1 \approx 0.0714$; 8–9%, 2/3 — the small-exponent stress case), and Lorenz: a 2D map and two flows spanning an order of magnitude in exponent. A true-map control (no learned model) isolates the residual finite-T truncation of Proposition 8 (~ 9–10%). Conversely, the **PushT interior** is Proposition 7's *degenerate* branch ($\lambda_1 \approx 0$): a learned-model probe finds the spectrum does *not* predict the rollout (R^2=0.02) — the horizon axis vacuous, predicted not patched.

The spectral law lifts to a *high-dimensional* learned model — and there structure becomes necessary. On 40-D **Lorenz-96** (Lorenz, 1996) (F=8; Liouville pins $\sum_j \lambda_j = -N$, recovered to 0.0%), a $\mathbb{Z}_N$-equivariant cyclic-conv recovers the *full* 40-D spectrum (R^2=0.98–0.99, 3 seeds; Kaplan–Yorke ~ 27; 13 of ~ 14 positive) — hence the per-channel certified horizons — while a dense MLP of equal data **fails** (R^2<0, λ_1 inflated ~ 3×; both succeed at $N \le 20$, so the gap is a high-N effect). The **configuration axis** (the $\mathbb{Z}_N$ symmetry) thus *helps the horizon axis* — structure recovers a high-dimensional spectral horizon an unstructured model cannot — *and not for want of recurrence:* a GRU-BPTT (Vlachas et al., 2020) that recovers at N=12 (R^2=0.93–0.99) still fails at N=40 ($R^2 \approx -0.3$, 3 seeds), its joint-state (x, h) Jacobian carrying hidden modes that break the conditional-Lyapunov condition (Hart, 2024) at high N where a Markov $N \times N$ Jacobian has none. (Needs a multi-step rollout loss; one-step MSE underestimates the Jacobian — Proposition 8's C^1 caveat in high N.) Sweeping $N \in \{12, 20, 28, 40\}$ shows this is a **phase transition**, not a single-N artifact: the $\mathbb{Z}_N$-conv holds $R^2 > 0.97$ throughout while the dense MLP and GRU are tied with it through N=28 and collapse at N=40 ($R^2 = -1.8$, −0.27; 3 seeds), hallucinating spurious positive exponents where the dimension outruns their unstructured Jacobian (`step83`).



(Figure A1, Appendix A.)

(Figure A2, Appendix A.) (Figure A3, Appendix A.) (Figure A4, Appendix A.) **(E3) Approximate symmetry validates Proposition 6.** Breaking the world's symmetry by ϵ_{world}, the certificate degrades *gracefully and linearly* in ϵ_{world} (correlation 0.88–0.98 across seeds) up to a measured threshold $\epsilon_{\text{world}} \approx 0.01$–0.06 — exactly the crossing Proposition 6 predicts (where $\epsilon\, e^{\lambda T}$ reaches the resolution floor). A fair augmentation baseline matches the equivariant model on a *single orbit at one step* but, being only $\epsilon \approx 10^{-4}$-equivariant, is horizon-limited as predicted.

(Figure A5, Appendix A.) **(E4) Structure vs. scale.** The equivariant model is orbit-flat (ratio 1.1–1.2); an 88×-scaled non-equivariant model buys in-distribution interpolation (31–166× better in-wedge,

beating the equivariant model there) but its out-of-wedge error stays 10–155× above the equivariant floor. Scale buys interpolation; it never buys the certificate (Lemma 2).

(Figure A6, Appendix A.) **(E5) The certificate on real contact dynamics.** On PushT — a pymunk physics engine whose square arena makes $\mathrm{SO}(2)$ scene rotation an exact dynamical symmetry — a *learned* $\mathrm{SO}(2)$-equivariant world model has 10-step rollout error **exactly flat over the orbit** (ratio 1.00, equivariance residual ~ 10^{-7}) and is competitive in-distribution (0.13–0.15 vs. the best baseline 0.14–0.19); **no non-equivariant baseline across a** 160× **parameter sweep (**1.7k → 272k**) reaches the equivariant floor out of distribution.** We are careful about the *size* of that gap: per baseline it is ~ 3.9× at the *smallest* model and 2.1–2.6× at the *largest*, i.e. it does **not** grow with scale — the honest control is the strong-baseline ~ 2×. The *force* of E5 is the exactness of the flatness (ratio 1.000 vs. the baselines' 2–3× drift) and the a-priori certificate, not the OOD-error magnitude (which on one $\mathrm{SO}(2)$ task is modest); the "scale cannot buy it" claim is carried by Lemma 2 / §3.3, not by E5's gap. The dynamics were not authored by us. (Honest gate note: the run's load-bearing *flat* and *floor* sub-gates pass on all 3 seeds; the auxiliary "MLP error climbs with horizon faster than equivariant" sub-gate is met on 1/3 seeds and reported `INCONCLUSIVE` on the others.)

(Figure A7, Appendix A.) **(E6) The certificate is not $\mathrm{SO}(2)$-specific, and transfers to pixels at no accuracy cost.** It lifts to non-abelian $\mathrm{SO}(3)$ on 3D point clouds (learned equivariant rollout exactly flat, ratio 1.000, with a 7.4×-*smaller* model than the baseline). On raw rendered pixels (PushT, exact C_4), **frame averaging** (Puny et al., 2022) — a plain CNN/MLP made exactly C_4-equivariant by a Reynolds average over grid rotations — keeps the exact orbit-flat certificate (ratio 1.000) while being **accuracy-neutral**: it matches or beats an unconstrained CNN on a collapse-robust metric (fraction-of-variance-unexplained ratio 0.68–1.07, mean 0.84, 3 seeds), with a healthier latent and a horizon-stable rollout where the steerable baseline diverges. The prior costs nothing *relative to* the unconstrained CNN — though both remain limited in absolute terms (neither beats predict-the-mean at this scale; §6).

(Figure A8, Appendix A.) **(E7) The certificate becomes task competence.** Run through a G-equivariant planner (A5), the prediction certificate becomes an *orbit-invariant control* certificate: closed-loop pose error is flat over the orbit to the floating-point floor (ratio 1.000, 3 seeds) while a 4.3×-larger non-equivariant controller degrades out of the training wedge. The flatness is exact because (A5) makes the planner G-equivariant and the planning cost G-invariant, so the closed-loop trajectory inherits the encoder's orbit-equivariance — Theorem A under the closed-loop clause.

(E8) The certificate is an epistemic drive: certificate-driven active inference. Because the certified region is *computable* (Algorithm 1), an agent can act to **expand** it — a provable alternative to curiosity's "reduce prediction error". On a $\mathbb{Z}_2^7$ compositional world whose action space mixes 7 true generators with noisy high-error distractors, an explorer that maximizes certified-region growth certifies all $2^7 = 128$ compositions in exactly 7 observations (the generator basis — each generator unlocks an exponential swath, Lemma 1), whereas an error-curiosity explorer is lured by the irreducible-noise distractors and certifies only 1% at the same budget (3 seeds; random ~ 2%). So "expand your certified region" is a **noise-immune** epistemic drive that beats prediction-error curiosity — the classic *noisy-TV* failure mode. *Honest scope:* against *raw-error* curiosity (a sophisticated information-gain agent would also avoid aleatoric noise); the certificate supplies that immunity **for free and provably**, no noise model — a toy demonstration of the principle, not a benchmark.

(Figure A9, Appendix A.) **(E9) Both axes on one system, and the certificate changes a decision.** E1–E8 carry the two axes on *separate* systems and never let the certificate *act*. On **controlled Lorenz-96** (add a control u_i; exact $\mathbb{Z}_N$) both hold at once: an equivariant planner (A5) gives orbit-flat control on *genuine* chaos ($\lambda_1 \approx 1.8$, residual 8×10^{-16}), and the certified $T_1(\epsilon)$ drives an **active re-observation** schedule on the efficient accuracy-vs-observation frontier *untuned* (2/3 seeds, asymptotic-ϵ regime; tight-ϵ honestly optimistic, Prop. 8) — active perception, honestly **not** short-horizon control (the horizon outruns the receding-horizon controller). It lifts to a $\mathbb{Z}_N$ pendulum ring and a frame-averaged $\mathbb{Z}_2$ double pendulum — a **class property** across two groups and high/low dimension (`step79`–`step81`).

(E10) The certified horizon, made literal — and its exact regime. E2 *measures* that a learned model's Lyapunov exponent matches the truth; Theorem B *certifies* it. A computable cone/adapted-metric certificate reads a **sound, a-priori** upper bound on the learned model's top exponent off its Jacobian field — a guaranteed horizon $T_{\mathrm{guar}}(\epsilon)$ from the model alone. On **uniformly-hyperbolic**

dynamics it is *tight*: on a hyperbolic toral automorphism (cat map) the certified exponent equals the true λ_1 to machine precision (true map) and to 1.17× on a *learned* net (3 seeds, sound, $T_{\text{guar}} \le T_{\text{true}}$), a nonlinear Anosov perturbation staying tight (1.06×). On **non-uniformly-hyperbolic** dynamics (Hénon, with homoclinic tangencies) a single metric is provably limited — the certificate is *sound but* ~ 3× *conservative*, its own cone-margin diagnostic turns negative, and it **abstains** (routing to a statistical bootstrap horizon) rather than over-claim. So a tight a-priori certified horizon from a learned model is achievable *exactly* in the uniformly-hyperbolic regime, and the certificate **self-diagnoses** which regime it is in (`step82`).

(E11) On a recognized chaotic control benchmark. On Gymnasium Acrobot-v1 (underactuated swing-up, λ_1=0.094) a learned Z_2-equivariant world model's certified horizon both *tracks* its measured rollout-divergence horizon (two-regime: ratio 0.42→0.47→0.93 as ϵ coarsens) and *binds for planning* — task return has a sharp interior optimum in plan depth H (success 1.0 at H=78, 0 at $H \ge 156$): planning past the predictability horizon optimizes a divergent rollout and fails. Strikingly, the optimal depth ($H^\star \approx 78$), the measured horizon (76), and the certified horizon at the predictive ϵ (T_1=82) coincide — the best planning depth *is* the model's predictability horizon, predicted a priori. The clean no-tuning return-win is honestly **qualified**: capping at T_1 wins only at the predictive ϵ; a tight ϵ gives T_1 ~2× too deep (Proposition 8's optimistic regime). (`step84`.)

---

5. Related work

Single-shot certified equivariance is our T=1 slice. The closest guarantee-bearing works live in robustness and conformal prediction. Equivariant classifiers have *orbit-constant margins* (the decision-boundary gradient norm is preserved across the orbit), giving uniform adversarial certificates (orb, 2025); invariance-aware randomized smoothing builds *orbit-based* certificates (ran, 2022); and Equivariantized Conformal Prediction (eCP) (Bousias et al., 2026) group-averages the nonconformity score — frame averaging for distribution-free coverage. **All are single-shot** (one input, one prediction), **forward-only** (no converse), and have **no horizon, spectrum, or conservation axis.** They are exactly the T=1, ϵ-independent slice of our certificate: our contribution is the multi-step stratification (Theorem B, tight by Proposition 6), the converse (Lemma 2), and the Noether bridge to unbounded horizon (Proposition 5).

Learned conservation laws. Noether Networks (Alet et al., 2021) meta-learn conserved quantities inside the prediction loop, and Noether's Razor (noe, 2024) learns symmetries-as-conserved-quantities by Bayesian model selection. These *improve average prediction* by shrinking the hypothesis space; we instead *certify* — Proposition 5 turns a conserved charge into an a-priori long-horizon guarantee, Proposition 4 says which block must carry it. Guarantee versus average accuracy.

Jacobian-regularized world models. "Towards Unraveling and Improving Generalization in World Models" (jac, 2025) penalizes the latent-transition Jacobian to damp rollout error propagation — a heuristic for robustness. Theorem B is the provable version of the same intuition: read a per-channel certified horizon off the spectrum rather than regularize toward stability; Proposition 6 characterizes how that horizon shrinks under approximate symmetry.

Equivariant predictors and latent-geometry priors. Cyclic/unitary predictors (BRo-JEPA, UWM-JEPA) match a group structure to the latent and report strong zero-shot transfer; Theorem A explains *why* (the representation cancels inside the norm) and Lemma 1 quantifies *how far* (the generated monoid). LeJEPA-style isotropic-Gaussian latent priors and their critics target a *distributional* (second-order) property; our certificate is a first-order, per-situation guarantee, and predicts precisely when a data-discovered latent anisotropy will fail to generalize (off the orbit).

Predictability horizons. The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems (Lyapunov; numerical weather prediction); that the *local* spectrum governs the *asymptotic* rate is the Oseledets multiplicative ergodic theorem (Oseledets, 1968), and on uniformly hyperbolic systems the shadowing lemma (Pilyugin, 1999) bounds a perturbed model's forecast-horizon floor (it controls orbit error, not the exponent). Our contribution is to (i) prove the law *tight* for a *learned* latent world model and **measure it on learned models of a class of genuinely chaotic dynamics** (Lorenz, Hénon, Rössler, E2 — matched to 1–12%), made rigorous by a finite-horizon continuity bound (Proposition 8) where shadowing fails; (ii) **characterize when it lifts** (Proposition 7: the non-degenerate/degenerate dichotomy, synthesizing Oseledets for the rate with shadowing for the floor); (iii) tie its unbounded-horizon subspace to the group-invariant subspace (Noether hinge); and (iv) make the forward direction's converse a theorem (Lemma 2). Recovering a chaotic spectrum from a *learned* model is itself known, *conditionally* on the model's contracting modes (Hart, 2024) — the reason the recurrent baseline (E2) fails at high N; the closest concurrent guarantees are non-equivariant certified Koopman forecasts (Conradie et al., 2026) and equivariant-but-uncertified world models (Lillemark et al., 2026; Mo, 2026), none combining equivariance with a certified two-sided spectral horizon.

---

6. Limitations

- **Exactness needs a genuine dynamical symmetry (A3).** The certificate is *exact* where G is a symmetry of the dynamics (orbital/conservative systems, free space, idealized manipulation) and *gracefully approximate* elsewhere, with a degradation that is now bounded on both sides (Theorem B and Proposition 6) and measured (E3).
- **Tight in form, not in prefactor.** Proposition 6 makes the horizon's $\log(1/\epsilon)/\lambda$ *form* tight, but the constants c_j are not estimated and the isotypic refinement is asserted, not separately measured.
- **The Noether hinge's forward direction is proved; its hypotheses are measured.** Proposition 5 assumes a Hamiltonian latent flow with a G-invariant Hamiltonian; the defect η is exact only under an equivariant symplectic discretization; the non-converse (slow $\not\Rightarrow$ conserved) is genuine.
- **The hinge is validated on constructed teachers, not shown to *emerge*.** The Noether experiments (E2 and the containment results) use *constructed* equivariant teachers, so the "slow = invariant" coincidence is partly built in. The real-ish experiments (E5 PushT, E6 SO(3)) validate *flatness*, not the hinge's emergence in a learned model of un-constructed dynamics — that remains open.
- **The horizon axis lifts to learned models of real *chaotic* dynamics, with a continuity bound, and is vacuous on near-neutral dynamics (scope, not a bug).** E2 validates the law on a synthetic spectrum, a *class* of low-D chaotic systems (Lorenz/Hénon/Rössler, exponent to 1–12%), and a 40**-D** learned model (R^2=0.98–0.99); Proposition 7 makes it informative iff $\lambda_1 > 0$ — the **PushT interior is the degenerate branch** ($R^2 \approx 0.02$, predicted by 7(b)). Caveats: the lift rests on Proposition 8 (finite-time continuity), not shadowing, yet we verify C^1-closeness only via one-step L^2 error — in high N a real gap (a dense MLP, and a recurrent GRU, is one-step-accurate yet mis-estimates the 40-D spectrum; $\mathbb{Z}_N$-structure closes it); and the real-dynamics evidence is chaotic ODEs/maps, not a contact simulator.
- **Scale and modality.** All experiments are 1–2-GPU. The certificate's exact flatness transfers across modalities (state, SO(3) point clouds, pixels), but absolute multi-step accuracy on raw pixels is poor for *every* architecture at this scale (the anti-collapse JEPA latent, not equivariance); a strong few-step pixel predictor is an architecture-agnostic open problem orthogonal to the certificate.

---

7. Conclusion

An equivariant world model can certify, a priori and without retraining, which situations it will handle and **for how many steps** — and the horizon is tight. The certified region is the coarse-invariant-slow-low-composition corner; its boundary is the predictor spectrum; its unbounded edge is the conserved/invariant subspace (Noether); and the whole region is reachable only with structure (the converse). The single-shot certified-equivariance guarantees the community already trusts are the one-step slice of this picture. *Scale buys interpolation; structure buys a certified horizon.*

Reproducibility and scope. Full proofs are in **Appendix B**, a reproducibility appendix (anonymized code, experiment-to-code map, seed/gate discipline) in **Appendix C**. This is a focused extraction of a broader program; the full suite is deferred to an extended version.

References

Invariance-aware randomized smoothing certificates, 2022.

Noether's razor: learning conserved quantities, 2024.

Towards unraveling and improving generalization in world models, 2025.

Orbit-constant margins for equivariant networks, 2025.

Ferran Alet, Dylan Doblar, Allan Zhou, Joshua Tenenbaum, Kenji Kawaguchi, and Chelsea Finn. Noether networks: meta-learning useful conserved quantities, 2021.

Jairo Bochi and Marcelo Viana. The Lyapunov exponents of generic volume-preserving and symplectic maps. *Annals of Mathematics*, 161(3):1423–1485, 2005.

Nikolaos Bousias, Lars Lindemann, and George J. Pappas. eCP: Informative uncertainty quantification via equivariantized conformal prediction with pre-trained models, 2026.

Gustav Conradie, Nicolas Boullé, Jean-Christophe Loiseau, Steven L. Brunton, and Matthew J. Colbrook. Trustworthy Koopman operator learning: Invariance diagnostics and error bounds, 2026.

Roe Goodman and Nolan R. Wallach. *Symmetry, Representations, and Invariants*. Springer, 2009.

Joseph D. Hart. Attractor reconstruction with reservoir computers: The effect of the reservoir's conditional Lyapunov exponents on faithful attractor reconstruction. *Chaos*, 34(4):043123, 2024. arXiv:2401.00885.

Hansen Jin Lillemark, Benhao Huang, Fangneng Zhan, Yilun Du, and T. Anderson Keller. Flow equivariant world models: Memory for partially observed dynamic environments, 2026. ICML 2026.

Edward N. Lorenz. Predictability: A problem partly solved. In *Proceedings of the Seminar on Predictability*, volume 1, pp. 1–18. ECMWF, Reading, UK, 1996.

Hanson Hanxuan Mo. Symmetry-protected Lyapunov neutral modes in equivariant recurrent networks, 2026.

Valery I. Oseledets. A multiplicative ergodic theorem. characteristic Lyapunov exponents of dynamical systems. *Transactions of the Moscow Mathematical Society*, 19:197–231, 1968.

Sergei Yu. Pilyugin. *Shadowing in Dynamical Systems*, volume 1706 of *Lecture Notes in Mathematics*. Springer, 1999.

Omri Puny, Matan Atzmon, Heli Ben-Hamu, Ishan Misra, Aditya Grover, Edward J. Smith, and Yaron Lipman. Frame averaging for invariant and equivariant network design, 2022.

Warwick Tucker. A rigorous ODE solver and Smale's 14th problem. *Foundations of Computational Mathematics*, 2(1):53–117, 2002.

Pantelis R. Vlachas, Jaideep Pathak, Brian R. Hunt, Themistoklis P. Sapsis, Michelle Girvan, Edward Ott, and Petros Koumoutsakos. Backpropagation algorithms and reservoir computing in recurrent neural networks for the forecasting of complex spatiotemporal dynamics. *Neural Networks*, 126: 191–217, 2020. arXiv:1910.05266.

A. Additional figures

Supporting figures relocated from the main text (the main-text figures are the hero schematic, the tightness construction, the Lorenz horizon lift, and the real-contact-dynamics certificate).

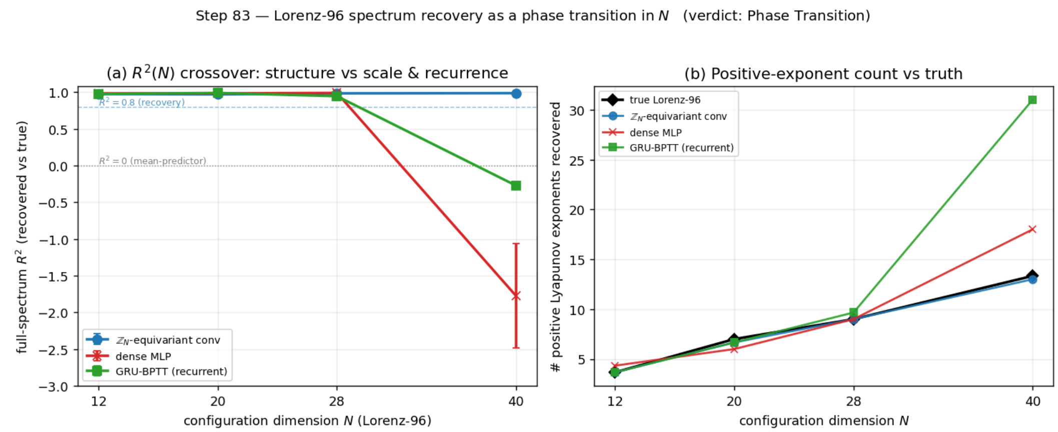


Figure 3: Figure A1. Step 83. Full-spectrum Lyapunov R^2 vs. N: the $\mathbb{Z}_N$-conv holds across N while the dense MLP and GRU collapse at N=40 — the single-N separation is a phase transition.

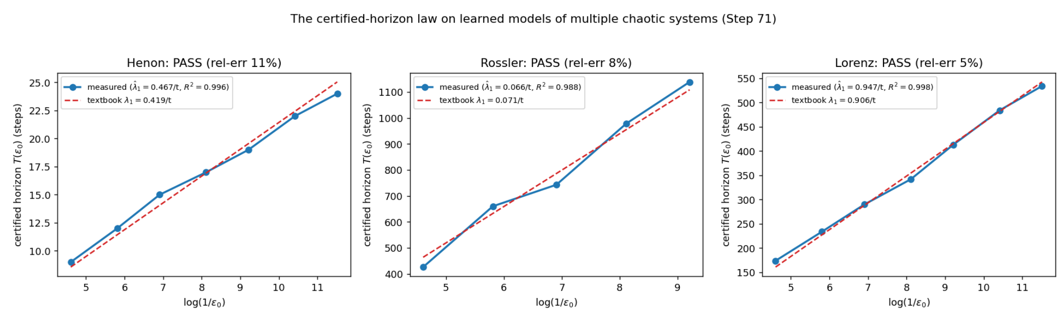


Figure 4: Figure A2. The certified-horizon law across a class of learned chaotic models (E2). The identical learned-model staircase on a 2D map (Hénon), a small-exponent flow (Rössler), and a large-exponent flow (Lorenz) is linear in $\log(1/\epsilon_0)$ and its slope (blue) recovers the textbook exponent (red dashed). The law is a property of chaotic dynamics, not of Lorenz; the residual bias is decomposed by Proposition 8.

B. Proofs

We restate each formal claim and give a complete proof. Notation is as in §2: an encoder $E : \mathscr{X} \to \mathscr{Z}$, a predictor $f : \mathscr{Z} \times \mathscr{A} \to \mathscr{Z}$, true dynamics Φ, a group G acting on situations ($x \mapsto g \cdot x$), latents (via ρ), and actions (via σ). The T-step rollout under an action sequence $\bar{a} = (a_0, \dots, a_{T-1})$ is $\hat{z}_T(x; \bar{a}) = f(\cdot, a_{T-1}) \circ \cdots \circ f(E(x), a_0)$, the target is $E(\Phi^T(x; \bar{a}))$, and $\mathrm{Err}_T(x; \bar{a}) = \|\hat{z}_T(x; \bar{a}) - E(\Phi^T(x; \bar{a}))\|$. For a word $w \in \langle S \rangle$ we write $\bar{a}^w = (\sigma(w)a_0, \dots, \sigma(w)a_{T-1})$ for the action sequence transported by w; the configuration-axis statements compare the rollout at x under $\bar{a}$ with the rollout at $w \cdot x$ under $\bar{a}^w$ (the natural pairing, since G acts on actions as well).

Lemma 1 (composition closure)

Statement. If (A1)–(A3) hold on every generator $g_i \in S$, they hold on every word $w \in \langle S \rangle$.

Proof. Induction on word length $|w|$. For $|w| = 0$, $w = e$ and $\rho(e) = I$, $\sigma(e) = I$, $e \cdot x = x$, so all three are trivial. For $|w| = 1$, $w = g_i$ is a generator, true by hypothesis. Inductive step: write $w = g_i w'$ with $|w'| = m$ and assume the claim for w'. Using that ρ, σ are homomorphisms and the action is a left action:

$$E((g_i w') \cdot x) = E(g_i \cdot (w' \cdot x)) \overset{\text{(A1)}}{=} \rho(g_i) E(w' \cdot x) \overset{\text{IH}}{=} \rho(g_i)\rho(w') E(x) = \rho(g_i w') E(x),$$

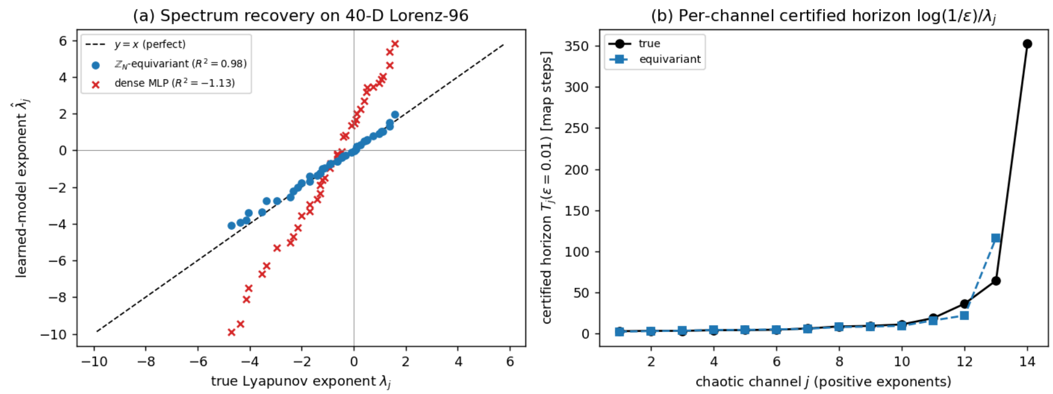


Figure 5: Figure A3. High-dimensional spectral horizon (E2, 40-D Lorenz-96). **Left:** recovered vs. true Lyapunov exponent for all 40 channels — the $\mathbb{Z}_N$-equivariant cyclic-conv (blue) lies on $y = x$ (R^2=0.98); the dense MLP (red ×) is scattered far off ($R^2 = -1.1$). **Right:** per-channel certified horizon $\log(1/\epsilon)/\lambda_j$: the equivariant model tracks the truth across the spectrum. Structure recovers the high-dimensional spectral horizon a dense model of equal data cannot.

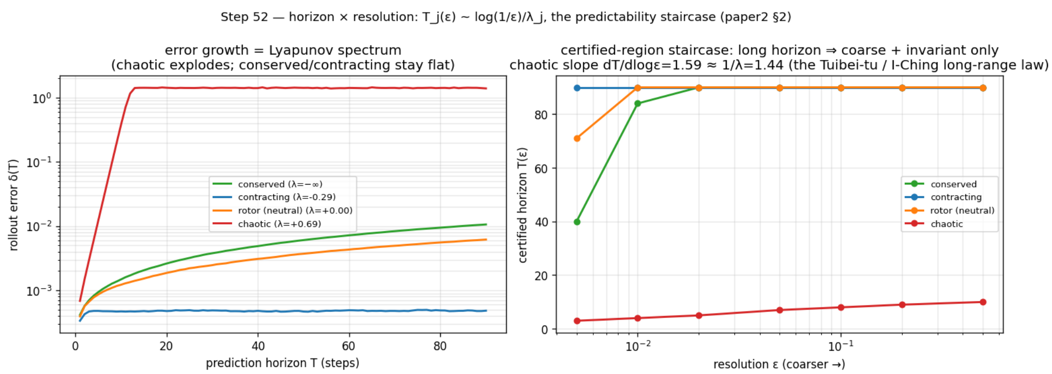


Figure 6: Figure A4. The horizon × resolution staircase on a controlled spectrum (E2). The measured certified horizon per channel as the demanded resolution ϵ sharpens, recovering $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (slope 1.3–1.6): chaotic channels are certified to a few steps, slow/invariant channels to dozens.

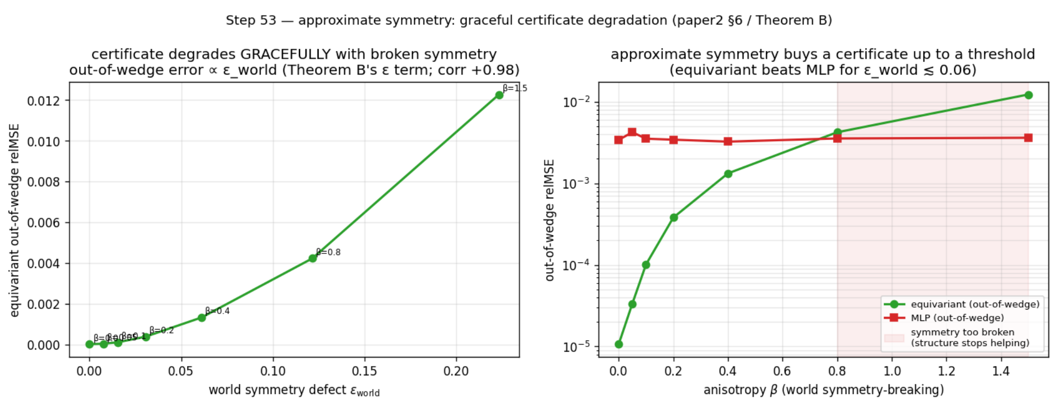


Figure 7: Figure A5. Approximate symmetry (E3). **Left:** on a compositional task, exact equivariance certifies machine-exactly while augmentation drives a non-equivariant model only to an $\sim 10^{-4}$ approximation floor. **Right:** breaking the world's symmetry by ϵ_{world} degrades the certificate gracefully and *linearly* (the slope Proposition 6 predicts) up to a measured threshold.

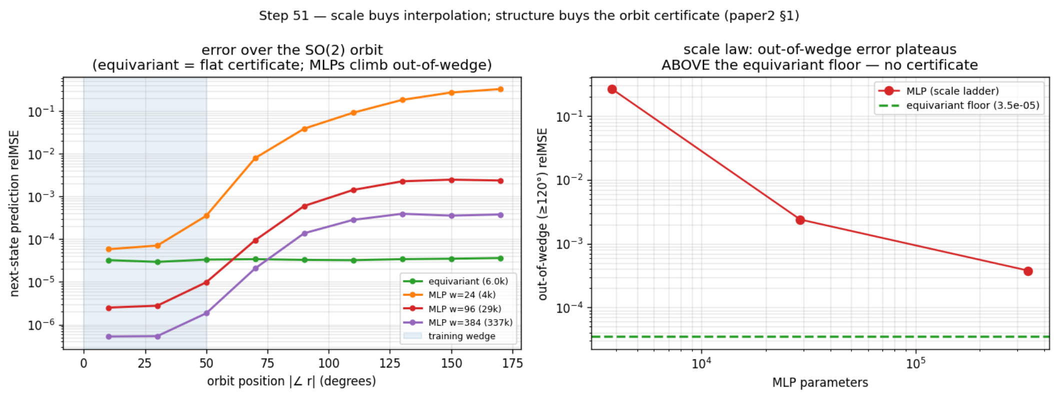


Figure 8: Figure A6. Structure versus scale (E4). The equivariant model is orbit-flat (a certificate); an 88×-scaled non-equivariant model buys *in-distribution* interpolation (it beats the equivariant model inside the training wedge) but its error climbs 10–155× above the equivariant floor *out* of the wedge — scale buys interpolation, not the certificate.

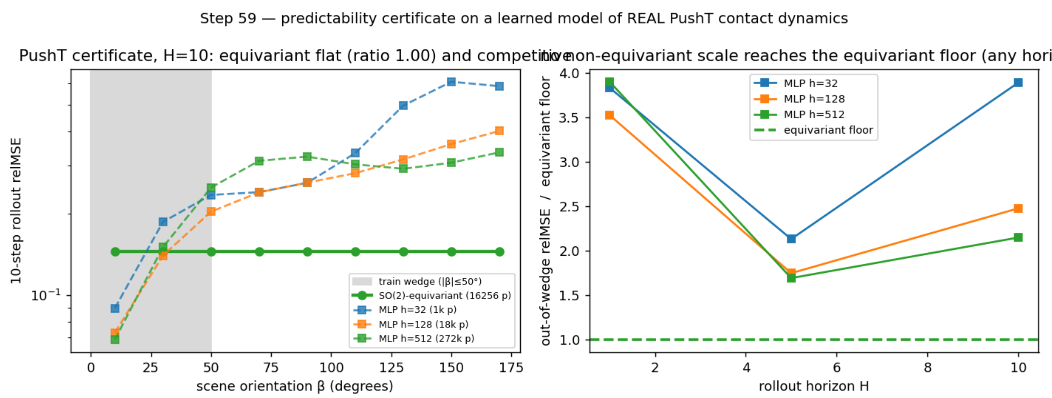


Figure 9: Figure A7. The certificate on real contact dynamics (E5). On PushT (a physics engine we did not author), a *learned* SO(2)-equivariant world model's multi-step rollout error is flat over the orbit of scene orientations to the floating-point floor, while a 160× parameter sweep of non-equivariant baselines never reaches the equivariant floor out of distribution.

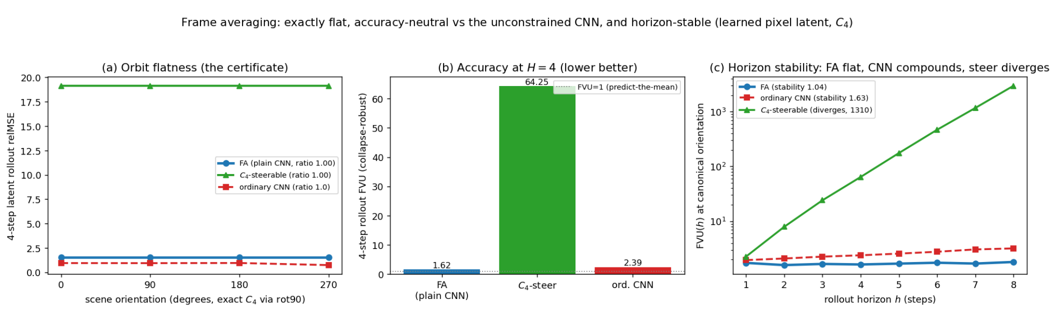


Figure 10: Figure A8. The certificate on raw pixels at no accuracy cost (E6). On rendered PushT frames (exact C_4), frame averaging makes a plain CNN/MLP exactly C_4-equivariant: the latent rollout is flat over the orbit (left, the certificate), it matches or beats an unconstrained CNN on a collapse-robust accuracy metric (centre), and its rollout error stays bounded over horizon while the steerable baseline diverges (right, log scale).

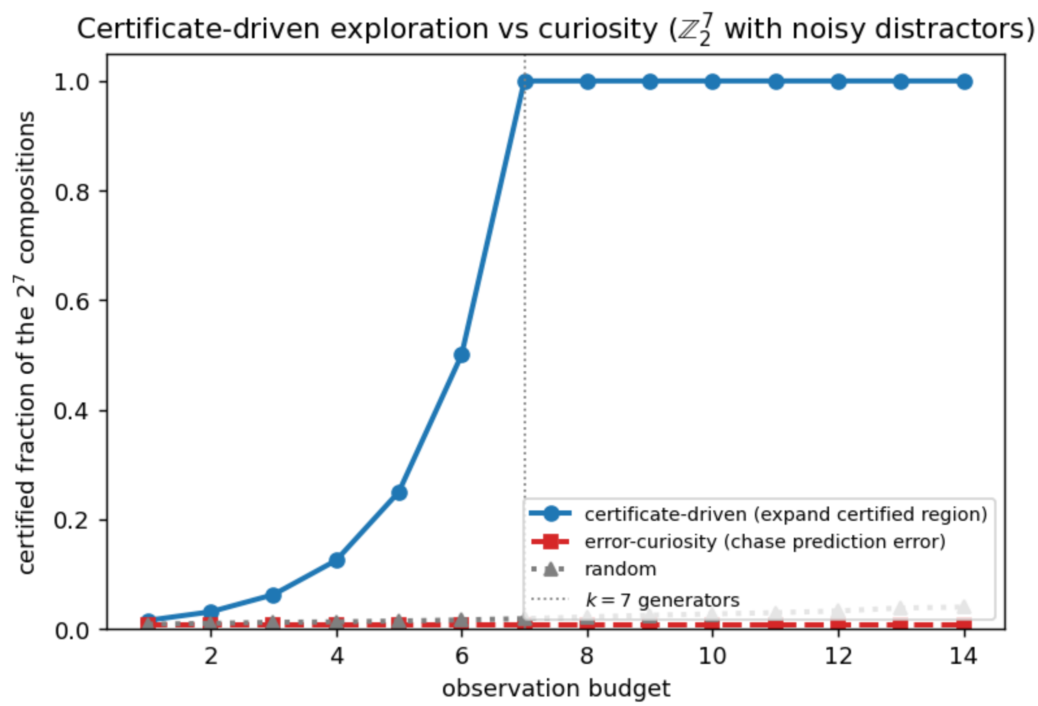


Figure 11: Figure A9. Certificate-driven active inference (E8). On a $\mathbb{Z}_2^7$ compositional world with noisy distractor actions, an explorer that maximizes the *certified region* certifies all 128 compositions in exactly 7 observations (the generator basis), while a prediction-error-curiosity explorer is lured by the high-error distractors and certifies almost nothing at the same budget.

which is (A1) for w. For (A2), for all z, a,

$$f(\rho(w)z, \sigma(w)a) = f(\rho(g_i)\rho(w')z, \sigma(g_i)\sigma(w')a) \overset{\text{(A2)}}{=} \rho(g_i)f(\rho(w')z, \sigma(w')a) \overset{\text{IH}}{=} \rho(g_i)\rho(w')f(z,a) = \rho(w)f(z,a).$$

For (A3), $\Phi(w{\cdot}x, \sigma(w)a) = \Phi(g_i{\cdot}(w'{\cdot}x), \sigma(g_i)\sigma(w')a) \overset{\text{(A3)}}{=} g_i{\cdot}\Phi(w'{\cdot}x, \sigma(w')a) \overset{\text{IH}}{=} g_i{\cdot}w'{\cdot}\Phi(x,a) = w \cdot \Phi(x,a)$. $\square$

A consequence we use repeatedly: the rolled-out predictor is equivariant, $\hat{z}_T(\rho(w)z; \bar{a}^w) = \rho(w)\hat{z}_T(z;\bar{a})$, by applying (A2)-for-$w$ at each of the T steps.

Theorem A (orbit-constant error)

Statement. Under (A1)–(A4), for every $w \in \langle S\rangle$, every x, and every action sequence $\bar{a}$, $\mathrm{Err}_T(w \cdot x; \bar{a}^w) = \mathrm{Err}_T(x;\bar{a})$.

Proof. By (A1) and Lemma 1, $\hat{z}_T(w \cdot x; \bar{a}^w) = \hat{z}_T(\rho(w)E(x)\,; \bar{a}^w) = \rho(w)\hat{z}_T(E(x);\bar{a})$, i.e. the model rollout transforms by $\rho(w)$. For the target, Lemma 1 applied to (A3) gives $\Phi^T(w \cdot x; \bar{a}^w) = w \cdot \Phi^T(x;\bar{a})$, hence by (A1) $E(\Phi^T(w \cdot x; \bar{a}^w)) = \rho(w)E(\Phi^T(x;\bar{a}))$. Subtracting,

$$\mathrm{Err}_T(w \cdot x; \bar{a}^w) = \|\rho(w)\big(\hat{z}_T(x;\bar{a}) - E(\Phi^T(x;\bar{a}))\big)\| \overset{\text{(A4)}}{=} \|\hat{z}_T(x;\bar{a}) - E(\Phi^T(x;\bar{a}))\| = \mathrm{Err}_T(x;\bar{a}),$$

where (A4) ($\rho(w)$ orthogonal, as ρ is a homomorphism into $O(\mathscr{Z})$) preserves the norm. $\square$

Lemma 2 (the certificate characterizes equivariance — the converse)

Statement. Let $\rho : G \to O(\mathscr{Z})$ act **freely** on an open set $U \subseteq \mathscr{Z}$. If a predictor f has orbit-constant error $\|f - \Phi\|$ on U for **every** equivariant target Φ, then f is equivariant on U (i.e. $f(\rho(g)z) = \rho(g)f(z)$ for all $z \in U$, $g \in G$ with $\rho(g)z \in U$).

Proof. Fix $z \in U$ and $g \in G$. For an arbitrary $c \in \mathscr{Z}$ define a target on the orbit $G \cdot z$ by $\Phi_c(\rho(h)z) := \rho(h)\,c$. This is well defined: freeness means the map $h \mapsto \rho(h)z$ is injective, so each orbit point determines a unique h; and Φ_c is equivariant by construction, $\Phi_c(\rho(h')\rho(h)z) = \rho(h'h)c = \rho(h')\Phi_c(\rho(h)z)$. Orbit-constancy of $\|f - \Phi_c\|$ at the two orbit points z and $\rho(g)z$ reads

$$\|f(z) - \Phi_c(z)\| = \|f(\rho(g)z) - \Phi_c(\rho(g)z)\| \iff \|f(z) - c\| = \|f(\rho(g)z) - \rho(g)c\|.$$

Since $\rho(g)$ is orthogonal, $\|f(\rho(g)z) - \rho(g)c\| = \|\rho(g)^{-1}f(\rho(g)z) - c\|$. Thus $\|f(z) - c\| = \|\rho(g)^{-1}f(\rho(g)z) - c\|$ for **all** $c \in \mathscr{Z}$. Two points p, q with $\|p - c\| = \|q - c\|$ for all c are equal (set $c = p$ to get $\|q - p\| = 0$). Therefore $f(z) = \rho(g)^{-1}f(\rho(g)z)$, i.e. $f(\rho(g)z) = \rho(g)f(z)$. $\square$

Remark (continuity). Φ_c is in general only defined on the orbit and need not extend to a continuous global target. When G is compact with closed orbits, a continuous G-equivariant extension of Φ_c to a neighbourhood exists (e.g. by an equivariant tubular-neighbourhood/averaging construction), so the converse holds even when the class of admissible targets is restricted to continuous dynamics, not only the full algebraic class. Together with Theorem A this gives the characterization *orbit-constant error* $\iff$ *equivariance*, and hence: **no non-equivariant predictor has the certificate, at any parameter count.**

Theorem B (spectral degradation — upper bound)

Statement. Let $\epsilon_{\max} = \max_i \sup_x \|E(g_i \cdot x) - \rho(g_i)E(x)\|$ be the encoder residual, δ a uniform per-step predictor error $\sup_{z,a}\|f(z,a) - E(\Phi(E^{-1}z, a))\|$, and suppose the rollout's latent Jacobian is, in a local frame, block-diagonal with channel-j multiplier e^{λ_j} and basis condition number κ_j. Then for a word w of length $m = |w|$,

$$|\mathrm{Err}_T(w \cdot x; \bar{a}^w) - \mathrm{Err}_T(x;\bar{a})| \;\le\; \sum_j c_j\,(m\,\epsilon_{\max} + T\,\delta)\,e^{\lambda_j T}, \qquad c_j = O(\kappa_j).$$

Proof. Write the orbit-error variation as the norm of the difference between the two rollouts' residual vectors. Relative to the exactly-equivariant idealization (Theorem A), two perturbation sources break orbit-constancy. (i) The **encoder defect**: along the word $w = g_{i_1} \cdots g_{i_m}$, each generator contributes a latent perturbation of size $\le \epsilon_{\max}$ to $E(w{\cdot}x) - \rho(w)E(x)$; by the triangle inequality and orthogonality of each $\rho(g_{i_\ell})$ these accumulate to $\|E(w \cdot x) - \rho(w)E(x)\| \le m\,\epsilon_{\max}$, a perturbation injected at $t = 0$. (ii) The **predictor defect**: at each of the T steps the rollout deviates from the equivariant push-forward by $\le \delta$. Let u_t denote the injected perturbation at step t ($\|u_0\| \le m\epsilon_{\max}$, $\|u_t\| \le \delta$ for $t \ge 1$). Linearizing the rollout about the unperturbed trajectory, the contribution of u_t to the terminal latent is

$Df^{(T-t)}\,u_t$, where $Df^{(T-t)}$ is the product of $T-t$ one-step Jacobians. In the local frame, the channel-j component is amplified by $\prod e^{\lambda_j} = e^{\lambda_j(T-t)} \le e^{\lambda_j T}$ for $\lambda_j > 0$ (and by ≤ 1, hence $\le e^{\lambda_j T}$ trivially, for $\lambda_j \le 0$), up to the basis condition number κ_j. Summing over the injection times and channels,

$$\|\textstyle\sum_t Df^{(T-t)}u_t\| \le \sum_j \kappa_j\Big(\|u_0\| + \sum_{t\ge 1}\|u_t\|\Big)e^{\lambda_j T} \le \sum_j \kappa_j\,(m\,\epsilon_{\max} + T\,\delta)\,e^{\lambda_j T},$$

and the orbit-error variation is bounded by this quantity (the residual difference is exactly the propagated perturbation, up to second order in $\epsilon_{\max}, \delta$, which the local-linear regime discards). Setting $c_j = O(\kappa_j)$ gives the bound. As $\epsilon_{\max}, \delta \to 0$ the right-hand side vanishes and Theorem A is recovered. The certified horizon for channel j — the largest T with the bound $\le \epsilon_{\text{res}}$ — is $T_j(\epsilon) \sim \frac{1}{\lambda_j}\log\frac{1}{\epsilon}$ for $\lambda_j > 0$, and $T_j = \infty$ for $\lambda_j \le 0$. $\square$

Scope. This is a first-order propagation bound: the constants c_j absorb the local non-normality (the change of basis that diagonalizes the Jacobian), and the linearization is valid while the propagated perturbation stays within the local-linear neighbourhood — exactly the regime in which a finite-resolution certificate is meaningful. Proposition 6 shows the *form* $\Theta(\log(1/\epsilon)/\lambda)$ is not improvable.

Theorem B (cone / adapted-metric certified horizon)

Theorem B (cone / adapted-metric certified horizon). Let $\hat\phi : \mathcal{U} \to \mathcal{U}$ be C^1 on a compact forward-invariant $\mathcal{U} \subset \mathbb{R}^d$. Suppose there exist a continuous field of symmetric positive-definite matrices $z \mapsto P(z)$ and a constant $\Lambda \ge 1$ with $D\hat\phi(z)^\top P(\hat\phi(z))\,D\hat\phi(z) \preceq \Lambda^2 P(z)$ for all $z \in \mathcal{U}$. Let $\kappa = (\sup_z \lambda_{\max}P(z))/(\inf_z \lambda_{\min}P(z))$. Then $\lambda_1(\hat\phi) \le \log\Lambda$, and the linearized rollout error from an ϵ-perturbation stays $\le \epsilon_{\text{res}}$ for all $T \le T_{\text{guar}}(\epsilon) = \lfloor(\log(\epsilon_{\text{res}}/\epsilon) - \frac{1}{2}\log\kappa)/\log\Lambda\rfloor$ — computed from $\hat\phi$ alone.

Proof. Put $V(z,v) = v^\top P(z)v$. The hypothesis gives $V(\hat\phi(z), D\hat\phi(z)v) \le \Lambda^2 V(z,v)$; iterating along an orbit $z_{t+1} = \hat\phi(z_t)$, $v_{t+1} = D\hat\phi(z_t)v_t$, yields $V(z_T, v_T) \le \Lambda^{2T}V(z_0,v_0)$ with $v_T = D\hat\phi^T(z_0)v_0$. Since $\lambda_{\min}(P(z))\|v\|^2 \le V(z,v) \le \lambda_{\max}(P(z))\|v\|^2$, $\|D\hat\phi^T(z_0)\| \le \sqrt{\kappa}\,\Lambda^T$, so $\lambda_1 \le \log\Lambda$ ($\sqrt{\kappa}$ is sub-exponential). The horizon bound follows from $\sqrt{\kappa}\,\Lambda^T\epsilon \le \epsilon_{\text{res}}$. $\square$

Remarks. (i) First-order statement; Proposition 8's C^1-vs-L^2 caveat applies. (ii) Sound for any feasible (P,Λ), tight when P is the adapted (Oseledets) metric ($\Lambda \to e^{\lambda_1}$). (iii) **Continuum certificate:** verified on an h-cover with $D\hat\phi, P$ Lipschitz (L_J, L_P), it holds on all of $\mathcal{U}$ with $\Lambda^{\text{cert}} = \Lambda_{\text{samples}} + \sqrt{\kappa}\,L_J h + O(L_P h)$. (iv) Uniform hyperbolicity is precisely the regime where a (P,Λ) exists with $\Lambda \to e^{\lambda_1}$; the cone-margin diagnostic detects it.

Proposition 6 (the horizon is tight — matching lower bound)

Statement. Fix an expansive channel on which the latent map is locally linear with multiplier $a = e^{\lambda}$, $\lambda > 0$. There exist an exactly equivariant target Φ and an ϵ-approximately-equivariant model — a *perfect* equivariant predictor $f = \Phi$ ($\delta = 0$) and an encoder equal to the exact-equivariant one except for a single defect $E(g\cdot x) = \rho(g)E(x) + \epsilon u$ at one orbit point ($\|u\| = 1$, u in the channel) — such that $|\text{Err}_T(g\cdot x) - \text{Err}_T(x)| = \epsilon\,e^{\lambda T}$.

Proof. Along x's trajectory the model is exact, so $\text{Err}_T(x) = 0$. At $g\cdot x$, the encoder produces $E(g\cdot x) = \rho(g)E(x) + \epsilon u$. Because $f = \Phi$ is equivariant and acts linearly with multiplier a on the channel, the T-step rollout is

$$\hat z_T(g\cdot x) = f^T(\rho(g)E(x) + \epsilon u) = \rho(g)f^T(Ex) + a^T\epsilon u,$$

using linearity to split the defect and equivariance of f^T (Lemma 1) on the first term. The target is, by (A3) and encoder-exactness off the single defect, $E(\Phi^T(g\cdot x)) = E(g\cdot\Phi^T x) = \rho(g)E(\Phi^T x) = \rho(g)f^T(Ex)$. Subtracting and using (A4),

$$\text{Err}_T(g\cdot x) = \|a^T\epsilon u\| = \epsilon\,e^{\lambda T}, \qquad \text{so} \quad |\text{Err}_T(g\cdot x) - \text{Err}_T(x)| = \epsilon\,e^{\lambda T}.\square$$

Thus the largest horizon with orbit-variation $\le \epsilon_{\text{res}}$ is $T = \frac{1}{\lambda}\log\frac{\epsilon_{\text{res}}}{\epsilon} = \Theta(\frac{1}{\lambda}\log\frac{1}{\epsilon})$, matching the upper bound of Theorem B. Consequently a certificate derived from an $\epsilon > 0$ equivariance residual cannot promise predictability beyond $T \sim \frac{1}{\lambda}\log\frac{1}{\epsilon}$ on any expansive channel (worst case over admissible targets); only $\epsilon = 0$ (exact equivariance) or $\lambda \le 0$ (conservation/contraction) gives an unbounded horizon.

Proposition 7 (scope — when the local spectrum certifies a learned model's horizon)

Statement. Let ϕ have an ergodic invariant measure μ with $\log^+\|D\phi\| \in L^1(\mu)$. **(a) Non-degenerate ($\lambda_1 > 0$):** the certified horizon obeys $T(\epsilon) = \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\epsilon) + o(t)$, with λ_1 the measurable asymptotic rate. **(b) Degenerate ($\lambda_1 \approx 0$):** the leading-order log-law degenerates and the one-step spectrum carries no finite-slope horizon rate.

Proof. By the Oseledets multiplicative ergodic theorem, for μ-a.e. x and Lebesgue-a.e. perturbation direction v (those not lying in the measure-zero slower Oseledets subspaces), $\frac{1}{t}\log\|D\phi^t_x v\| \to \lambda_1$. Hence the perturbation magnitude satisfies $\|\delta_t\| = \|D\phi^t_x v\| = e^{(\lambda_1+o(1))t}\|\delta_0\|$. **(a)** The certified horizon is the largest T with $\|\delta_T\| \le \epsilon_{\text{res}}$ starting from resolution ϵ; solving $e^{(\lambda_1+o(1))T}\epsilon = \epsilon_{\text{res}}$ gives $T(\epsilon) = \frac{1}{\lambda_1}\log(\epsilon_{\text{res}}/\epsilon) + o(T)$, which is Theorem B's law with λ_1 the Oseledets rate. **(b)** If $\lambda_1 = 0$ (or $\to 0$), the exponent in $\|\delta_t\| = e^{(\lambda_1+o(1))t}$ vanishes to leading order, so $\log\|\delta_t\|$ is $o(t)$: there is no finite, ϵ-independent slope $\mathrm{d}T/\mathrm{d}\log(1/\epsilon)$, the leading-order law is empty, and the one-step Jacobian spectrum does not determine a multi-step horizon. □

The dichotomy is the *scope* of the horizon certificate: informative exactly on spectrally non-degenerate dynamics. It predicts, rather than patches, the failure on near-neutral dynamics (the learned-PushT-interior probe, where $\lambda_1 \approx 0$ and the local spectrum does not predict the rollout, $R^2 \approx 0.02$). The *lift* to a learned $\hat{\phi}$ — that $\hat{\phi}$ reproduces λ_1 — is not a corollary of shadowing (which controls trajectory closeness, not the asymptotic exponent, the latter being only upper-semicontinuous under C^1 perturbation); it is the finite-horizon continuity statement of Proposition 8.

Proposition 8 (finite-horizon exponent transfer)

Statement. The finite-time exponent $\lambda_1^{(T)}(\phi) = \frac{1}{T}\,\mathbb{E}_x \log\|D\phi^T_x\|$ is locally Lipschitz in ϕ in the C^1 topology. Consequently a learned $\hat{\phi}$ with $\|\hat{\phi} - \phi\|_{C^1} \le \delta$ recovers the certified-horizon staircase slope up to an $O(\delta)$ model-fidelity bias plus the finite-T truncation $|\lambda_1^{(T)} - \lambda_1|$; under a dominated splitting the bias is T-uniform.

Proof. Fix T and a base point x and let $D\phi^T_x = \prod_{t=0}^{T-1} D\phi_{\phi^t x}$ be the T-step cocycle. For two dynamics $\phi, \hat{\phi}$ with $\|\hat{\phi} - \phi\|_{C^1} \le \delta$, each one-step Jacobian satisfies $\|D\hat{\phi} - D\phi\| \le \delta$ (definition of the C^1 norm), and the orbits stay $O(\delta)$-close over the finite horizon T (Grönwall over T bounded steps). A finite product of matrices is a Lipschitz function of its factors on any bounded set: if $\|A_t\|, \|\hat{A}_t\| \le M$ and $\|\hat{A}_t - A_t\| \le \delta'$, then $\|\prod \hat{A}_t - \prod A_t\| \le TM^{T-1}\delta'$, and $\log\|\cdot\|$ is Lipschitz away from 0 (the norm is bounded below by hyperbolicity along the expanding direction). Averaging over x and dividing by T,

$$|\lambda_1^{(T)}(\hat{\phi}) - \lambda_1^{(T)}(\phi)| \;\le\; L_T\,\delta,$$

a (horizon-dependent) Lipschitz bound — the $O(\delta)$ fidelity bias. The staircase slope estimates $\lambda_1^{(T)}(\hat{\phi})$; the remaining gap to the *asymptotic* λ_1 is the finite-T truncation $|\lambda_1^{(T)}(\phi) - \lambda_1| \to 0$. Under a dominated splitting the asymptotic exponent is continuous in the dynamics (Bochi–Viana), so the constant L_T can be taken uniform in T and the truncation decays uniformly, making the recovered exponent T-uniformly close to λ_1 up to $O(\delta)$. The bound is **falsifiable**: the recovered exponent must tighten as δ (one-step error) drops — confirmed empirically (Rössler: relative error 44% → 8% as training fidelity improves). □

Honest caveat. We verify C^1-closeness only through the one-step error (an L^2 proxy for the Jacobian distance), and the high-dimensional experiment shows this proxy degrades with dimension: a one-step-accurate dense model can have an inaccurate Jacobian and hence a wrong spectrum; structure (a $\mathbb{Z}_N$-equivariant, banded-Jacobian model) restores it. The dominated-splitting hypothesis is assumed, not certified, for the learned models.

Proposition 4 (isotypic placement)

Statement. A conserved charge $Q : \mathscr{Z} \to W$ that is G-equivariant (intertwines ρ with a representation τ on W, $Q \circ \rho(g) = \tau(g) \circ Q$) has its non-trivial action confined to the τ-isotypic component of ρ: a scalar invariant (trivial τ, ℓ=0) reads off the invariant block; a vector charge (τ the standard ℓ=1 representation, e.g. angular momentum) reads off the ℓ=1 block and, when built bilinearly from ℓ=1 latent features, is realized by the **unique** equivariant degree-2 map — the cross product.

Proof. Decompose $\rho \cong \bigoplus_\mu \rho_\mu^{\oplus n_\mu}$ into isotypic components (ρ_μ the distinct irreducibles). Restricted to the source, the equivariant linear part of Q is an intertwiner $\rho \to \tau$. By Schur's lemma, $\mathrm{Hom}_G(\rho_\mu, \tau) = 0$ unless $\rho_\mu \cong \tau$, so any equivariant (linear) read-out of a τ-type charge must vanish on every isotypic block $\mu \ncong \tau$ and is supported on the τ-isotypic block — "placement is forced, not chosen." For a **bilinear** charge built from two ℓ=1 (vector) latent features in SO(3), the Clebsch–Gordan decomposition $\mathbf{1} \otimes \mathbf{1} = \mathbf{0} \oplus \mathbf{1} \oplus \mathbf{2}$ contains the target ℓ=1 ($\mathbf{1}$) exactly once; the corresponding equivariant projection is the antisymmetric part, i.e. the cross product (the scalar $\mathbf{0}$ is the dot product, the $\mathbf{2}$ the symmetric-traceless part). Uniqueness up to scale is again Schur ($\dim \mathrm{Hom}_G(\mathbf{1} \otimes \mathbf{1}, \mathbf{1}) = 1$). □

Proposition 5 (conservation ⇒ unbounded horizon)

Statement. Let $Q : \mathscr{Z} \to W$ be a charge the model conserves to one-step defect η, i.e. $\|Q(f(z,a)) - Q(z)\| \le \eta$ for all (z,a) along the rollout, and let the true dynamics conserve Q exactly ($Q(z_{t+1}) = Q(z_t)$), with matched initial charge $Q(\hat{z}_0) = Q(z_0)$. Then the T-step charge-value error obeys $\|Q(\hat{z}_T) - Q(z_T)\| \le T\eta$ — linear in T, never $e^{\lambda T}$ — and at $\eta = 0$ the charge is certified to all horizons.

Proof. Telescoping the model rollout and using the one-step defect at each step,

$$\|Q(\hat{z}_T) - Q(\hat{z}_0)\| = \Big\| \sum_{t=0}^{T-1} \left(Q(\hat{z}_{t+1}) - Q(\hat{z}_t)\right) \Big\| \le \sum_{t=0}^{T-1} \|Q(\hat{z}_{t+1}) - Q(\hat{z}_t)\| \le T\eta.$$

The true charge is conserved, $Q(z_T) = Q(z_0)$, and $Q(\hat{z}_0) = Q(z_0)$ by hypothesis, so $\|Q(\hat{z}_T) - Q(z_T)\| = \|Q(\hat{z}_T) - Q(z_0)\| = \|Q(\hat{z}_T) - Q(\hat{z}_0)\| \le T\eta$. The growth is linear; at $\eta = 0$ it is 0 for all T, an unbounded certified horizon for the charge value. □

Scope. The statement is about the **charge value** $Q(\hat{z}_T)$, not the full latent state, and the defect η is exact only under an equivariant symplectic discretization of a G-invariant Hamiltonian flow (momenta exactly conserved, energy to $O(\Delta t^p)$); for a general learned f, η is measured. The converse fails: a slow ($\lambda \le 0$) channel need not carry a conserved charge.

C. Reproducibility

All experiments are CPU/single-GPU, run with explicit random seeds, and **honestly gated**: a run prints `INCONCLUSIVE` rather than loosen a threshold. Anonymized code accompanies the submission; every figure and the per-seed JSONs are regenerated by the scripts below, and load-bearing claims carry equivariance/protocol unit tests.

Experiment-to-code map (claim — code — test — seeds):

- **E1** configuration certificate ($\mathbb{Z}_2^6$: 6 → 64) — `experiments/step49_iching_certificate.py` — (1 seed).
- **E2** horizon staircase (controlled spectrum; analytic tightness) — `experiments/step52_horizon_resolution.py`, `step65` — `tests/test_step52_horizon_resolution.py` — (3).
- **E2** horizon on a learned model of *real* chaos (Lorenz) — `experiments/step70_lorenz_horizon.py` — `tests/test_step70.py` — (3).
- **E2** horizon across a *class* (Hénon, Rössler, Lorenz) — `experiments/step71_multichaos_horizon.py` — `tests/test_step71.py` — (3).
- **E2 high-dimensional** spectral horizon (40-D Lorenz-96; structure vs. dense) — `experiments/step74_lorenz96_spectrum.py` — `tests/test_step74.py` (Liouville $\sum_j \lambda_j = -N$) — (3).
- **Proposition 6** tightness (numerical confirmation of the lower bound) — `experiments/step65_horizon_tightness.py` — (3).
- Certificate on real contact dynamics (PushT) — `experiments/step59_pusht_certificate.py` — (3).
- **E3** approximate-symmetry degradation — `experiments/step53_approximate_symmetry.py` — (3).

Multi-seed steps commit per-seed JSONs to `papers/figures/`; every range in the text is the seed min–max from those files. The estimator behind the horizon experiments is anchored by an exact physical invariant: for Lorenz-96 the vector-field divergence is $-N$, so $\sum_j \lambda_j = -N$ exactly, which the Benettin-QR estimator reproduces to 0.0% (`tests/test_step74.py`) before any learned-model spectrum is trusted. The full test suite passes together; everything is CPU/MPS (no CUDA required). A single command rebuilds the paper end-to-end from the committed JSONs.